Affine Dynamics and Spectral Compression of Decimal Mountain Concatenations

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Abstract

For $M(k) = 1 \parallel 2 \parallel \cdots \parallel k \parallel (k-1) \parallel \cdots \parallel 1$, we prove that every prime p with $\mathrm{ord}_p(10) = 7 \cdot 2^a 5^c$ or $\mathrm{ord}_p(10) = 11 \cdot 3^b 2^a 5^c$, $a, c \geq 0$, $0 \leq b \leq 2$, divides at least one $M(k)$. The argument first compresses the stable affine drift for a *fixed* coprime-order component v to at most $\ell = \mathrm{lcm}(t, \mathrm{ord}_t(10))$ cyclotomic cosets, where $t = v/\gcd(v, 9)$, uniformly over arbitrarily large powers of $2, 5$. Both drift coordinates become polynomials of degree at most two in $m = r/\gcd(r, \ell)$ over a fixed-degree cyclotomic field. For $v = 11, 33, 99$, an explicit invariant proves nonzero drift algebraically when the order is odd; ten fixed norms and a complete finite sieve settle the even branch. We also give a direct witness whenever the $2, 5$-part of the order divides 20, with no restriction on its coprime part and no nonzero-drift assumption. This does not resolve the unrestricted OEIS A077187 conjecture.

1 Affine drift and the period reduction

The primes 2 and 5 never divide a mountain, whose final digit is 1. The primes 3 and 11 have witnesses $M(3) = 12321$ and $M(2) = 121$. Unless specified otherwise, the spectral formulas below are for $p \geq 7$. Set $r = \mathrm{ord}_p(10) = uv$ with $u = 2^a 5^c$, $(v, 10) = 1$. Write $C_k = 1 \parallel \cdots \parallel k$, $Y_k = 10^{-D(k)}(k \parallel \cdots \parallel 1) \pmod p$, where $D(k)$ is the total digit count. For a complete d-digit layer, put

$$L = 10^{d-1}, \quad q = 10^d, \quad n = 9L, \quad A = q^n, \quad S = \sum_{i<n} q^i, \quad J = \sum_{i<n} iq^i.$$

The exact layer map is

$$(C, Y) \longmapsto \big(AC + (q-1)S - J,\ A^{-1}(Y + LS + J)\big).$$

We will prove that a stable block of T digit layers is a translation of displacement (U, V). Consequently its successive end states satisfy

$$(C_{10^{jT}-1}, Y_{10^{jT}-1}) = (C_{10^T-1}, Y_{10^T-1}) + (j-1)(U, V), \qquad j \geq 1.$$

If $(U, V) \neq (0, 0)$, the normalized residues of $M(10^{jT})$, $M(10^{jT} + 1)$ have affine slopes $10U + V$ and $1000U + V$. For $p \geq 7$, $p \neq 11$, these cannot both vanish, yielding a divisible mountain with $1 \leq j \leq p$.

Set $\delta = \gcd(v, 9)$, $t = v/\delta$, $b_0 = 9/\delta$, and $h = \mathrm{ord}_t(10)$, with $\mathrm{ord}_1(10) = 1$. Define

$$\ell = \mathrm{lcm}(t, h), \qquad g = \gcd(r, \ell), \qquad m = r/g, \qquad T = m\ell.$$

The key point is that ℓ depends only on v, not on a, c. Also $T \geq r \geq \max(a, c)$, so the block starting at $T + 1$ is in the stable range.

Lemma 1 (Reduced short-period phase)**.** *For every stable digit layer d, the diagonal multiplier $A_d = 10^{9d10^{d-1}}$ has period ℓ. Furthermore $\prod_{j=d}^{d+\ell-1} A_j = 1 \pmod p$. Thus its cumulative phase P_d, started at $d = T + 1$, also has period ℓ. The stable multipliers generate precisely the subgroup of order t.*

Proof. Take d large enough that $u \mid 10^{d-1}$. Let $\zeta = 10^m$, of order g, and let $\eta = u^{-1} \pmod t$. Since $t \mid g$, division of the integer exponent by $u\delta$ gives

$$A_d = \zeta^{(g/t) b_0 \eta\, d 10^{d-1}}.$$

For $t = 1$, read this as $A_d = 1$. Periodicity follows from $t \mid \ell$ and $10^\ell = 1 \pmod t$. For the product it suffices to prove $E_\ell = \sum_{j=1}^{\ell} j10^{j-1} = 0 \pmod t$. Away from 3 this follows from $81E_\ell = 1 + (9\ell - 1)10^\ell$. For the 3-part, induction gives $s \mid E_s$ when $s = 3^n$: as $10^s = 1 \pmod{3s}$,

$$E_{3s} = \sum_{k=0}^{2} 10^{ks} \left(E_s + ks \sum_{j=1}^{s} 10^{j-1} \right) \equiv 3E_s \equiv 0 \pmod{3s}.$$

Partitioning any multiple of s into length-s blocks gives the required divisibility. A shifted length-ℓ sum also vanishes: the extra geometric sum $(10^\ell - 1)/9$ is divisible by t, using $v_3(10^\ell - 1) = 2 + v_3(\ell)$ at 3. Finally $(b_0\eta, t) = 1$, so A_{T+1} has order t.

Since $r \mid T$, the parameters q, L repeat after T; the phases do too, and their product in $T = m\ell$ layers is 1. The layer maps therefore repeat, with a translation as their block map. This proves the asserted affine block identity. □

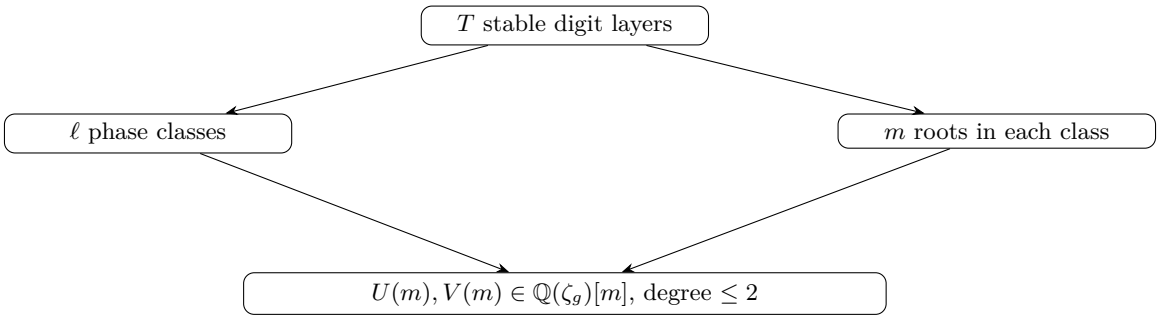


Figure 1: The unbounded stable block compresses to bounded-dimensional cyclotomic algebra once v is fixed.

2 Coset sums and quadratic spectral coordinates

Let $\zeta = 10^m \pmod p$, a primitive g-th root. For $j = 1, \ldots, \ell$, all layers $d = T + j + t\ell$ $(0 \leq t < m)$ share the same $A = A_j$ and $P = P_j$. Their values $q = 10^d$ are precisely the m roots of $z^m = c = \zeta^j$. Define

$$F_1 = \sum_{\substack{z^m = c \\ z \neq 1}} \frac{z}{z-1}, \qquad F_2 = \sum_{\substack{z^m = c \\ z \neq 1}} \frac{z}{(z-1)^2}.$$

Logarithmic differentiation of $z^m - c$, with $z = 1$ removed if necessary, gives

	F_1	F_2
$c \neq 1$	$mc/(c-1)$	$m^2c/(c-1)^2$
$c = 1$	$(m-1)/2$	$-(m^2-1)/12.$

For a nonsingular full layer, its ascending and descending block residues are

$$B = (A-1)\big(1 + q/(q-1)^2\big) - 9LA/(q-1),$$
$$R^* = (A-1)\big(L/(q-1) - q/(q-1)^2\big) + 9LA/(q-1).$$

The singular layer $q = 1$ contributes $B = R^* = 9/200$. Put $\alpha = P^{-1}(1 - A^{-1})$, $\beta = P(A-1)$, $n_0 = m - \mathbf{1}_{c=1}$.

Theorem 1 (Bounded-degree spectral compression)**.** *For each fixed v, the drift equals the sum of ℓ coset contributions*

$$\begin{aligned} U_j &= \alpha(n_0 + F_2) - \tfrac{9}{10}P^{-1}F_1 + \mathbf{1}_{c=1}\tfrac{9}{200}P^{-1}, \\ V_j &= \beta(F_1/10 - F_2) + \tfrac{9}{10}PAF_1 + \mathbf{1}_{c=1}\tfrac{9}{200}P. \end{aligned} \tag{1}$$

Moreover,

$$A_j = \zeta^{(g/t)b_0\eta\, j10^{j-1}}, \quad P_j = \prod_{s<j} A_s, \quad \eta = u^{-1} \pmod t,$$

so U, V are polynomials of degree at most two in m, with coefficients in the fixed cyclotomic field $\mathbb{Q}(\zeta_g)$, where $g \leq \ell$.

Proof. In the ascending long word the j-th layer block has weight $(PA)^{-1}$, while the descending block has weight P. Substitute the two block residues, sum over the m roots in each phase class, and include the singular contribution. The displayed formulas follow. Their only dependence on m is through m, m^2, while all phases are powers of ζ_g. □

Lemma 2 (Involutive degree drop for $\mathrm{ord}_v(10) \le 2$)**.** *If $v \mid 99$, then the coefficients of m^2 in both compressed drift coordinates vanish identically: $U_2 = V_2 = 0$ in the cyclotomic coefficient field.*

Proof. For $v = 1$, all stable multipliers equal 1. Otherwise $10^2 \equiv 1 \bmod v$ implies $A_{-j} = A_j^{-1}$. Because $\prod_{j=1}^{\ell} A_j = 1$, this yields the palindrome $P_{\ell+1-j} = P_j$. Hence both phase differences $\alpha_j = P_j^{-1} - P_{j+1}^{-1}$ and $\beta_j = P_{j+1} - P_j$ change sign under $j \mapsto \ell - j$. The coefficient of m^2 in F_2 is the inversion-even function $f(c) = c/(c-1)^2$ for $c \neq 1$, extended by $f(1) = -1/12$. Thus $U_2 = \sum_j \alpha_j f(\zeta^j)$ and $V_2 = -\sum_j \beta_j f(\zeta^j)$ cancel pairwise under the involution. □

3 Explicit linear coordinates for coprime parts 11, 33, 99

Write

$$r = 11w, \quad w = 3^b 2^a 5^c, \quad 0 \le b \le 2, \qquad \gamma = w^{-1} \pmod{11}.$$

Lemma 1 gives $t = 11, \ell = 22$, with $g = 11$ if $a = 0$ and $g = 22$ otherwise. Set $\omega = \zeta^{g/11}$ and $\tau = \omega^{9\gamma}$. The phases satisfy

$$A_j = \omega^{9\gamma j(-1)^{j-1}}, \qquad P_{2n} = \tau^n, \qquad P_{2n+1} = \tau^{-n}.$$

In particular $\sum_{j=1}^{22} P_j = 0$. The factor 3^b must be included in γ; discarding it changes the actual phase.

Define $X(c) = c/(c-1)$ for $c \neq 1$, with $X(1) = 1/2$. Inversion gives $X(c^{-1}) = 1 - X(c)$. The phase palindrome and formula (1) yield

$$U_1 = -\frac{9}{10} \sum_j P_j^{-1} X(\zeta^j), \qquad V_1 = -\frac{11}{10} \sum_j P_j X(\zeta^j).$$

The finite Fourier sums needed to evaluate these expressions are

$$\sum_{c^{11}=1} c^A X(c) = 11/2 - A \quad (0 \le A \le 10),$$
$$\sum_{c^{11}=-1} c^A X(c) = 11/2 \qquad (0 \le A \le 10),$$
$$\sum_{c^{11}=-1} c^{-A} X(c) = -11/2 \qquad (1 \le A \le 10).$$

For the first identity, inverse pairing gives the value at $A = 0$, and each successive difference is -1, by the root power sums. For the second, logarithmic differentiation of $x^{11} + 1$ gives the initial value and the successive differences vanish. The third follows by multiplying by $c^{11} = -1$.

Theorem 2 (Odd branch: a nonzero algebraic invariant)**.** *Every prime with $r = 11 \cdot 3^b 5^c$, $0 \le b \le 2$, has nonzero drift.*

Proof. Here $g = 11$. Choose $A \in \{1, \dots, 10\}$ with $2A = 9\gamma \pmod{11}$, and put $z = \zeta^A$, $D = 11 - 2A$. The even phase indices have $P = c^A$, the odd ones $P = zc^{-A}$. The Fourier sums, together with the singular classes $j = 11, 22$, give

$$U = \frac{99}{200}(1 + z^{-1}) + \frac{9D}{20} m(1 - z^{-1}),$$
$$V = -\frac{81}{200}(1 + z) - \frac{11D}{20} m(1 - z).$$

Their linear terms cancel in

$$\boxed{11zU - 9V = \frac{909}{100}(z + 1).}$$

Since z has order 11, $z + 1 \neq 0$. The prime factors of $909 = 9 \cdot 101$ are inadmissible: the orders of 10 modulo 3 and 101 are 1 and 4. Thus the right side is nonzero. No norm certificate is needed in this branch. □

For $g = 22$, take instead $A = 9\gamma \pmod{11}$ in $\{1, \dots, 10\}$, with $z = \zeta^A$, $D = 11 - 2A$. Even and odd indices now traverse $c^{11} = 1$ and $c^{11} = -1$. The same sums and the single singular class give

$$U = \frac{99}{200} + \frac{9m}{20}(D - 11z^{-1}), \qquad V = -\frac{81}{200} - \frac{11m}{20}(D - 11z).$$

Consequently

$$4000z(U_1V_0 - U_0V_1) = 9(891 + 40Dz - 1331z^2). \tag{2}$$

This necessary common-root obstruction uses no division by either slope, so it remains valid when a slope vanishes modulo p. The order n_A of z is 22 for odd A, and 11 for even A. Define the ten fixed integers

$$N_\gamma = \left|\mathrm{Res}_x\big(9(891 + 40Dx - 1331x^2), \Phi_{n_A}(x)\big)\right|. \tag{3}$$

These are exactly the existing even-branch $v = 11$ norms; multiplying by the root of unity z does not change their absolute norm.

Theorem 3 (Even branch: complete exact certificate)**.** *Every prime with $r = 11 \cdot 3^b 2^a 5^c$, $0 \le b \le 2$, $a \ge 1, c \ge 0$, has nonzero drift.*

Proof. All ten integers (3) are nonzero, as checked by exact resultants in `certify_involutive_families.py`. Simultaneous zero drift would force $p \mid N_\gamma$, since (2) vanishes at a primitive n_A-th root in $\mathbb{F}_p$. Also

$$r < p \le N_\gamma, \qquad p \mid \Phi_r(10).$$

The script enumerates every (a, b, c) with $11 \cdot 3^b 2^a 5^c \le \max_\gamma N_\gamma$, and checks the actual phase-dependent bound $r \le N_\gamma$ at each triple. Modular evaluation uses

$$\Phi_r(X) = \Phi_{22\,3^{[b>0]}5^{[c>0]}}\left(X^{2^{a-1}3^{\max(b-1,0)}5^{\max(c-1,0)}}\right).$$

Every prime dividing the substitution exponent already divides the base index, so this identity follows by identifying the primitive roots. The base polynomial has degree at most 80. The complete computation gives

b	0	1	2
Admissible indices	3887	3796	3702
$\gcd(N_\gamma, \Phi_r(10))$	1	1	1

Each raw gcd and each count is asserted. Thus all 11385 possibilities are excluded. The bound comes from a necessary norm divisor condition; it is not a cutoff in the desired infinite family. □

Corollary 1. *Every prime whose order has coprime-to-10 part dividing 99 divides some $M(k)$. The branches $11 \mid v$, $a \ge 1$ are computer-assisted; the remaining branches have algebraic proofs.*

Proof. For $v \mid 9$, Lemma 1 gives $A = P = 1$ and $T = r$. Each regular layer contributes $U + V = 0$; the unique $q = 1$ layer contributes $9/100 \ne 0$. For $v = 11, 33, 99$, use the two theorems. The two-offset argument gives a witness with $1 \le j \le p$. The primes 3 and 11 were handled directly. □

4 The existing quadratic family with coprime part seven

For $v = 7$, $h = 6$, $\ell = 42$, and $g = 7$ if $a = 0$, otherwise $g = 14$. The exact coordinates $U(m), V(m)$ are quadratic in m. Their Sylvester resultant

$$\mathcal{R} = (U_2V_0 - U_0V_2)^2 - (U_2V_1 - U_1V_2)(U_1V_0 - U_0V_1)$$

has, after scaling by 40000, a nonzero exact norm, independent of the six phase values γ:

g	$\lvert\mathrm{Norm}(40000\mathcal{R})\rvert$
7	33476324853960868807109434619064591936
14	11497454360840474493325194486198045857885568 9

If simultaneous zero drift occurred in this family, its prime p would divide the corresponding fixed nonzero norm N, satisfy $r = \mathrm{ord}_p(10) \mid p - 1$ and $p \mid \Phi_r(10)$, hence $r < p \le N$. The exact scripts evaluate $\Phi_r(10) \bmod N$ using base cyclotomic polynomials of degree at most 24 and modular powering. For all $53 + 4479$ admissible indices in the $v = 7$ norms, they verify that $\gcd(N, \Phi_r(10))$ has no prime divisor outside r, by repeated gcd removal. Since a prime of exact order r cannot divide r, zero drift is impossible. The affine two-offset argument therefore proves existence for every prime of order $7 \cdot 2^a 5^c$, without dependence on older smooth-order certificates.

5 A direct witness when the smooth part divides 20

Theorem 4 (An arbitrary coprime part)**.** *If $u \mid 20$, there is a divisible mountain, regardless of whether the stable drift vanishes and without restricting v.*

Proof. Use the stable period T above. For every $d \geq 2$, $20 \mid 9d10^{d-1}$, so only the first layer can differ from its stable continuation. Put $\alpha = 10^9$, $\widehat{\alpha} = 10^{9(T+1)10^T} \pmod{p}$, and $\rho = \widehat{\alpha}/\alpha$. The first-layer map is

$$C' = \alpha(C + 10/81) - 91/81, \qquad Y' = \alpha^{-1}(Y + 1/81) + 80/81.$$

Let the stable block be the translation by $B = (U, V)$. Its remaining layers have diagonal multiplier $(\widehat{\alpha}^{-1}, \widehat{\alpha})$. Thus the actual first block ends at $D + B$, where

$$D = ((\rho^{-1} - 1)10/81,\ (\rho - 1)/81).$$

After p blocks the state is $D + pB = D$. Within a $q \neq 1$ window starting at L, the exact recurrence gives

$$\frac{M(L+n)}{10^{D(L+n-1)}} = qAq^n + B_0 q^{-n} - \frac{2q}{(q-1)^2}, \quad A = C + \frac{L}{q-1} + \frac{1}{(q-1)^2}, \quad B_0 = Y - \frac{L}{q-1} + \frac{q}{(q-1)^2}.$$

At the window starting at 10^{pT}, its modular parameters are $q = 10, L = 1$, giving the square

$$\frac{(10z - \rho)^2}{81\rho z}, \qquad z = 10^n.$$

Choose $n = (9 \cdot 10^T - 10) \bmod r$, with $0 \leq n < r$. Then $z = \rho/10$, so $k = 10^{pT} + n$ is a witness. The window is long enough since $r < p$. The divisions by 81 are valid for $p \neq 3$; the case 3 has the direct witness already given. □

6 The boundary of the short-period mechanism

Lemma 1 also contains the nine-torsion lifting lemma: for $v = sv_0$, $s \mid 9$, $(v_0, 30) = 1$, the reduced phase component is $t = v_0$, and its effective phase is $(su)^{-1}$. If $3 \nmid \ell_0$, the field degree is unchanged. The separate `certify_fixed_v33_v99.py` certificate checks 3850 and 3757 indices, including the odd branches now settled algebraically. Its 15 independent prime-field tests agree with the new formulas.

Lemma 3 (Exact 3-adic boundary)**.** *Suppose $r = \mathrm{ord}_p(10) = 11 \cdot 3^b 2^a 5^c$ with $b \geq 1$, and set $A_d = 10^{9d10^{d-1}}$ on stable digit layers. For $t \geq 0$,*

$$\boxed{\mathrm{ord}(A_{d+22\cdot 3^t}/A_d) = 3^{\max(b-2-t,0)}.}$$

Consequently the 22-period folding is valid precisely for $b \leq 2$ among these 3-power extensions.

Proof. Let $L = 22 \cdot 3^t$ and $e_d = 9d10^{d-1}$. The exact difference

$$e_{d+L} - e_d = 9 \cdot 10^{d-1}\big(L + (d+L)(10^L - 1)\big)$$

is divisible by 11 and by the smooth factor $2^a 5^c$ in the stable range. By the 3-adic lifting-the-exponent lemma, $v_3(10^L - 1) = 2 + t$, while $v_3(L) = t$. Therefore $v_3(e_{d+L} - e_d) = 2 + t$ exactly. The claimed order follows from $\mathrm{ord}_p(10) = 11 \cdot 3^b 2^a 5^c$. □

This is an obstruction to reusing the 22-phase proof, not an obstruction to the existence of divisible mountains.

7 Reproducibility and scope

The files `certify_fixed_v7.py` and `certify_involutive_families.py` generate the exact norms and check every phase-dependent cyclotomic gcd, without factoring the large norm integers or enumerating target primes. Independent prime-field tests are in `audit_fixed_v7.py` and `audit_fixed_v11.py`. The file `audit_reduced_spectral.py` compares the reduced phases and explicit cosets with an independent full-layer recurrence. These finite comparisons check implementation consistency; the universal identities follow from the proofs above. Dedicated CI jobs reproduce the certificates. The new linear formulas and exhaustive sieve also received a separate audit recorded in `audit_spectral_update_2026-10-18.md`. The independent nine-torsion checks are preserved in `certify_fixed_v33_v99.py` and `audit_fixed_v33_v99.py`.

The spectral theorem applies to arbitrary fixed v, but a nonzero resultant and the elimination of its actual-order prime factors are *not* established for arbitrary v. The general all-primes A077187 problem remains open.

The present proofs impose no restriction on v when $u \mid 20$, and no restriction on u in the certified coprime-part families. Outside their union, the unresolved question is either to prove nonzero drift or to force a subgroup root on an actual zero-drift orbit. The separate note `zero_drift_continuation_2026-10-18.md` reduces the first-window tail phase to order $u/\gcd(u, 20)$, but also shows why three fixed initial windows do not universally suffice. Those additional reachability results are not assumptions in any of the existence proofs in this paper.

References

[1] OEIS Foundation, *A077187*, https://oeis.org/A077187.

[2] Q. Wang, *Decimal mountain concatenation research notes and exact certificates*, https://github.com/dongxuelian2/oeis-papers/pull/8, 2026.